

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 6 Summer 2026 (Regular)

Course : Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCVC602 - FOUNDATION ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Which of the following is an undisturbed soil sampling method commonly used in clayey soils?
 - A) Split spoon sampler
 - B) Auger boring
 - C) Thin-walled Shelby tube sampler
 - D) Wash boring
- 2 Core recovery in rock drilling is defined as:
 - A) Ratio of core length to borehole depth
 - B) Percentage of intact core length obtained to the length drilled
 - C) Ratio of rock fragments to drilling time
 - D) Depth at which drilling stops
- 3 Which field test is most suitable for bearing capacity evaluation in cohesionless soils?
 - A) Vane shear test
 - B) Standard Penetration Test (SPT)
 - C) Plate load test only
 - D) Unconfined compression test
- 4 Terzaghi's bearing capacity theory assumes that the foundation is:
 - A) Deep and flexible
 - B) Shallow and rigid
 - C) Deep and rigid
 - D) Shallow and flexible
- 5 Skempton's bearing capacity factors are primarily applicable for:
 - A) Sands under drained condition
 - B) Clays under undrained condition
 - C) Silts with partial drainage
 - D) Gravels under seismic loading

- 6 Effect of water table near the foundation level mainly results in:
 - A) Increase in bearing capacity
 - B) No change in bearing capacity
 - C) Reduction in effective stress and bearing capacity
 - D) Increase in cohesion
- 7 Expansive soils are identified mainly by high values of:
 - A) Void ratio
 - B) Plasticity index
 - C) Permeability
 - D) Relative density
- 8 Swelling pressure in expansive soils increases with:
 - A) Increase in moisture content
 - B) Increase in surcharge load
 - C) Reduction in clay content
 - D) Increase in grain size
- 9 Negative skin friction in piles occurs when:
 - A) Pile settles more than surrounding soil
 - B) Soil settles more than pile
 - C) Pile is driven in rock
 - D) Load is applied eccentrically
- 10 Under-reamed piles are most suitable for foundations in:
 - A) Sandy soils
 - B) Expansive clay soils
 - C) Gravelly soils
 - D) Peat soils
- 11 Factor of safety in slope stability is defined as the ratio of:
 - A) Driving forces to resisting forces
 - B) Resisting forces to driving forces
 - C) Shear stress to shear strength
 - D) Normal stress to shear stress
- 12 For an infinite slope in purely cohesive soil, stability mainly depends on:
 - A) Angle of internal friction
 - B) Unit weight of soil
 - C) Cohesion and slope angle
 - D) Permeability

Q2. Solve the following.

- A) Differentiate between disturbed and undisturbed soil sampling and discuss their significance. 6
- B) Describe field tests used for bearing capacity evaluation and discuss their limitations. 6

Q3. Solve the following.

- A) Define the following terms:- 6
 - 1) Bearing capacity
 - 2) Net pressure intensity
 - 3) Ultimate bearing capacity
 - 4) Net ultimate bearing capacity
 - 5) Net safe bearing capacity
 - 6) Safe bearing capacity
- B) Discuss the different modes of bearing capacity failure with diagram. 6

- Q4. Solve Any Two of the following.**
- A) What are expansive soils? Explain in detail the characteristics of expansive soils? 6**
 - B) Define and explain the following terms: 6**
 - 1) Free Swell
 - 2) Swelling Potential
 - 3) Swelling Pressure
 - C) Define settlement of foundation. Explain the different types of settlement in shallow foundations. 6**
- Q5. Solve Any Two of the following.**
- A) A group of 9 piles with 3 piles in a row was driven into soft clay extending from 6 ground level to a great depth. The diameter and length of piles were 30 cm and 10 m respectively. The unconfined compressive strength of clay is 70 kN/m^2 . If the piles were spaced at 90 cm c/c , compute the allowable load on the pile group on the basis of shear failure criteria for a factor of safety of 2.5. Neglect bearing at the tip of piles. Take $m = 0.6$ for shear mobilization around each pile. 6**
 - B) Explain the sheet pile walls in detail using neat sketches. 6**
 - C) Draw a neat, labelled sketch of typical section of well foundation. Also, draw 6 neat sketches of common types of shapes of well foundations. 6**
- Q6. Solve Any Two of the following.**
- A) What are the different types of slopes. Explain in detail with neat sketch and 6 example? 6**
 - B) A slope is to be constructed at an inclination of 30° with the horizontal. Determine 6 the safe height of the slope at factor of safety of 1.5. The soil has the following properties: $c = 15 \text{ kN/m}^2$, $\phi = 22.5^\circ$ and $\gamma = 19 \text{ kN/m}^3$ 6**
 - C) Enlist and explain in detail the different types of slope failures? 6**

*** End ***